



May 2025

AM3322-1 – BIG DATA ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- 1. What is Big Data?**
A) A type of database
B) A large collection of structured and unstructured data
C) A programming language
D) A networking protocol
- 2. Which of the following is a characteristic of Big Data?**
A) Volume, Velocity, and Variety
B) Veracity and Visualization
C) Data Mining
D) Database Transactions
- 3. What is an example of unstructured Big Data?**
A) SQL Databases
B) CSV Files
C) Social Media Posts
D) Spreadsheets
- 4. Which company originally developed Hadoop?**
A) Microsoft
B) Google
C) Facebook
D) Yahoo
- 5. Which of the following is NOT a Big Data technology?**
A) Hadoop
B) Spark
C) Tableau
D) MongoDB
- 6. What does Hadoop primarily use for distributed storage?**
A) MySQL
B) HDFS
C) PostgreSQL
D) NoSQL
- 7. Which of the following is a Big Data processing framework?**
A) TensorFlow
B) Apache Spark
C) Django
D) React
- 8. What is a Data Warehouse?**
A) A relational database used for OLTP transactions
B) A system for storing and analyzing structured and semi-structured data
C) A temporary storage space for Big Data
D) A type of NoSQL database
- 9. What was the original purpose of Hadoop?**
A) To store images
B) To manage social media data
C) To support the Nutch search engine
D) To analyze stock market data
- 10. What are the two core components of Hadoop?**
A) HDFS and YARN
B) YARN and SQL
C) MapReduce and Spark
D) HDFS and MySQL
- 11. What does HDFS stand for?**
A) Hadoop Data File System
B) Hadoop Distributed File System
C) Hadoop Development Framework System
D) Hadoop Directory File Storage

12. What are the two phases of MapReduce?
 A) Mapping and Indexing
 B) Mapping and Reducing
 C) Transforming and Filtering
 D) Loading and Fetching
13. What is the output of the Map phase?
 A) Final results
 B) Intermediate key-value pairs
 C) Database tables
 D) Raw data
14. What is the function of the Reduce phase?
 A) Collects and summarizes intermediate key-value pairs.
 B) Maps data into key-value pairs
 C) Performs indexing
 D) Sorts unstructured data
15. In a Hadoop cluster, where does the MapReduce program execute?
 A) On a single machine
 B) Distributed across multiple nodes
 C) On the NameNode only
 D) Inside a relational database
16. What is the function of the Hive Driver?
 A) Manages HiveQL query execution
 B) Stores metadata
 C) Loads data into HDFS
 D) Provides an API for machine learning models
17. What is a Partition in Hive?
 A) A logical division of a table based on column values
 B) A type of NoSQL database
 C) A function used in MapReduce
 D) A component of the Hadoop cluster
18. Which of the following is a Data Definition Language (DDL) command in Hive?
 A) INSERT
 B) CREATE TABLE
 C) SELECT
 D) UPDATE
19. What is Apache Pig?
 A) A query language for structured databases
 B) A machine learning tool
 C) A scripting platform for processing large data sets
 D) A graph processing engine
20. What language does Pig use?
 A) SQL
 B) HiveQL
 C) MapReduce
 D) Pig Latin

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Big Data. Explain how three V's of bigdata is extended to four V's	8	L*2	1	1
	b) With a neat diagram explain data warehouse architecture.	8	L2	1	1
2.	a) Elaborate on the functionalities of namenode and datanode.	8	L2	2	1
	b) Explain HDFS commands with example: i)mkdir ii)cp iii)cls iv) copyToLocal	8	L2	2	1
3.	a) Explain YARN map reducing components of resource manager in detail	8	L3	1	2
	b) Explain scheduling and its types with an example.	8	L3	2	2
Unit – II					
4.	a) Explain in detail the different categories of NoSQL databases.	8	L3	3	2
	b) What is a key space in NoSQL databases, and how is it defined?	8	L2	3	1
5.	a) Write following MongoDB commands with example: i) create database ii) drop database iii) show database	8	L3	4	2
	b) Explain in detail how HBase can be represented in key –value and multi-map structure?	8	L2	4	1
6.	a) Explain data models of Casendra database.	8	L2	4	1
	b) Elaborate on the working of MapReduce module.	8	L2	3	1
Unit III					
7.	a) Describe hive data models.	8	L2	5	1

b) Write a note with an example on the following hive commands

- i) To create a database named "Class" with comments and database, properties.
- ii) To describe a database and extended database.
- iii) To display a list of all databases.
- iv) To alter the database properties
- v) To drop database.
- vi) To create managed table named 'STUDENT'
- vii) To create external table named 'EXT_STUDENT'.
- viii) To load data into the table from file named class.tsv.

8	L2	5	1
8	L3	5	1
8	L2	5	1

8. a) Explain the working of hive with examples.

b) Explain the workflow between Hive and Hadoop

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
